Time series: ARMA forecasting

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The homework is provided on the last slide

Forecasting ARMA processes

Since an ARMA process is a stationary process, we can apply the algorithm of forecasting stationary processes for forecasting ARMA processes, see Lecture 6.

Let $x_1, \ldots, x_n$ be a time series, h be a positive integer and $R(k)$ be a given covariance function.

1. Compute $\hat{\mu} = (x_1 + x_2 + \ldots + x_n)/n$.
2. Construct the time series $y_i = x_i - \hat{\mu},\ i = 1, \ldots, n$.
3. Compute the matrix $\Gamma = \Big(R(i-j)\Big)_{i,j=1}^{n}$ and the vector $b_h = (R(h+n-1), \ldots, R(h+1), R(h))^T$.
4. Compute the vector $\gamma_h = \Gamma^{-1} b_h$.
5. Compute $\hat{y}_{n+h} = \gamma_h^T Y$, where $Y = (y_1, y_2, \ldots, y_n)^T$.
6. Compute $\hat{x}_{n+h} = \hat{\mu} + \hat{y}_{n+h}$ and $\hat{\sigma}^2_{n+h;x} = R(0) - b_h^T \Gamma^{-1} b_h$.

Forecasting AR(1) processes

Task. Let $x_1, \ldots, x_n$ be a time series from an AR(1) process $x_t = ax_{t-1} + \epsilon_t$. Compute h-step ahead forecast.

From slide 5 of Lecture 9, we have

$$R(k) = \sigma^2 \frac{a^{|k|}}{1 - a^2}.$$

Then the equation $\Gamma\gamma_h = b_h$ has the form

$$\frac{\sigma^2}{1-a^2} \begin{pmatrix} 1 & a & \ldots & a^{n-1} \\ a & 1 & \ldots & a^{n-2} \\ \vdots & & & \\ a^{n-1} & a^{n-2} & \ldots & 1 \end{pmatrix} \begin{pmatrix} \gamma_{h1} \\ \gamma_{h2} \\ \vdots \\ \gamma_{hn} \end{pmatrix} = \frac{\sigma^2}{1-a^2} \begin{pmatrix} a^{n+h-1} \\ a^{n+h-2} \\ \vdots \\ a^h \end{pmatrix}.$$

The solution is $\gamma_h = (0, \ldots, 0, a^h)^T$.

Forecasting AR(1) processes

Now the predictor $\hat{x}_{n+h}$ has the form

$$\hat{x}_{n+h} = (x_1, \ldots, x_n)^T \gamma_h = 0 \cdot x_1 + 0 \cdot x_{n-1} + a^h x_n = a^h x_n.$$

and the variance of the predictor is

$$\hat{\sigma}^2_{n+h;x} = R(0) - b_h^T \Gamma^{-1} b_h = \frac{\sigma^2}{1-a^2} - a^{2h} \frac{\sigma^2}{1-a^2} = \sigma^2 \frac{1-a^{2h}}{1-a^2}.$$

Theorem. For a time series $x_1, \ldots, x_n$ from an AR(1) process $x_t = a x_{t-1} + \epsilon_t$, the h-step ahead predictor is

$$\hat{x}_{n+h} = a^h x_n,$$

and the MSE of $\hat{x}_{n+h}$ is $\mathrm{MSE}(\hat{x}_{n+h}) = \sigma^2 \frac{1-a^{2h}}{1-a^2}$. If ϵ_t is Gaussian, then the prediction bound for x_{n+h} is

$$\left(\hat{x}_{n+h} - z_{1-\alpha/2} \sigma \sqrt{\frac{1-a^{2h}}{1-a^2}}, \hat{x}_{n+h} + z_{1-\alpha/2} \sigma \sqrt{\frac{1-a^{2h}}{1-a^2}} \right).$$

Example on forecasting AR(1)

Task. Let $x_1, \ldots, x_{100}$ be a time series from an AR(1) process $x_t - \mu = a(x_{t-1} - \mu) + \epsilon_t$ with $\mu = 9$, $a = 0.6$, $\sigma^2 = 0.1^2$; and let $x_{100} = 8.9$. Compute the predictors $\hat{x}_{101}, \hat{x}_{102}, \hat{x}_{103}$ and prediction bounds.

The given process is causal since $|a| < 1$. Then we have

$$\hat{x}_{101} = \mu + a\,(x_{100} - \mu) = 9 + \ 0.6\ \ (8.9 - 9) = 8.94$$
$$\hat{x}_{102} = \mu + a^2(x_{100} - \mu) = 9 + \ 0.6^2\ (8.9 - 9) = 8.964$$
$$\hat{x}_{103} = \mu + a^3(x_{100} - \mu) = 9 + \ 0.6^3\ (8.9 - 9) = 8.9784.$$

The prediction bounds for x_{101}, x_{102}, x_{103} are

$$x_{101}: \ (8.94 - z_{1-\alpha/2}0.1 \ \ , \qquad 8.94 + z_{1-\alpha/2}0.1)$$
$$x_{102}: \ \left(8.964 - z_{1-\alpha/2}0.1\sqrt{\tfrac{1-0.6^4}{1-0.6^2}}, 8.964 + z_{1-\alpha/2}0.1\sqrt{\tfrac{1-0.6^4}{1-0.6^2}}\right)$$
$$x_{103}: \ \left(8.9784 - z_{1-\alpha/2}0.1\sqrt{\tfrac{1-0.6^6}{1-0.6^2}}, 8.9784 + z_{1-\alpha/2}0.1\sqrt{\tfrac{1-0.6^6}{1-0.6^2}}\right)$$

Forecasting a causal ARMA process

Let $x_1, \ldots, x_n$ be a time series from a causal ARMA(p, q) process $a(B)x_t = b(B)\epsilon_t$, i.e.

$$x_t - a_1 x_{t-1} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q}.$$

Since x_t is causal, there are the coefficients $c_0, c_1, c_2, \ldots$ such that elements of the time series can be represented as

$$x_n = \sum_{i=0}^{\infty} c_i \epsilon_{n-i}, \quad j = 1, 2, 3, \ldots,$$

and $c_0 = 1$, see slide 7 of Lecture 8 and slide 4 of Lecture 9. Now we want to construct the predictor $\hat{x}_{n+h}$ in the form

$$\hat{x}_{n+h} = \sum_{j=0}^{\infty} \tau_{h+j} \epsilon_{n-j}$$

where optimal τ_{h+i}, $i = 0, 1, 2, \ldots$ minimize the MSE.

MSE of predictor for ARMA process

The mean square error (MSE) of $\hat{x}_{n+h}$ is

$$\begin{aligned}
\mathbf{E}(x_{n+h}-\hat{x}_{n+h})^2 &= \mathbf{E}(\sum_{i=0}^{\infty} c_i\epsilon_{n+h-i} - \sum_{j=0}^{\infty} \tau_{h+j}\epsilon_{n-j})^2 \\
&= \mathbf{E}(\sum_{i=0}^{\infty} c_i\epsilon_{n+h-i} - \sum_{i=h}^{\infty} \tau_{i}\epsilon_{n+h-i})^2 \\
&= \mathbf{E}\left(\sum_{i=0}^{h-1} c_i\epsilon_{n+h-i} + \sum_{i=h}^{\infty} (c_i-\tau_i)\epsilon_{n+h-i}\right)^2 \\
&= \sum_{i=0}^{h-1} c_i^2\mathbf{E}\epsilon_{n+h-i}^2 + \sum_{i=h}^{\infty} (c_i-\tau_i)^2\mathbf{E}\epsilon_{n+h-i}^2 \\
&= \sigma^2\sum_{i=0}^{h-1} c_i^2 + \sigma^2\sum_{i=h}^{\infty} (c_i-\tau_i)^2.
\end{aligned}$$

The minimum of MSE is attained for $\tau_j = c_j$, $j = h, h+1, \ldots$.

Forecasting of ARMA processes via MA

Theorem. For a time series $x_1, \ldots, x_n$ from a causal ARMA(p, q) process $a(B)x_t = b(B)\epsilon_t$, which can be written as $x_j = \sum_{i=0}^{\infty} c_i \epsilon_{j-i}$, where ϵ_t is a white noise process with variance σ^2, the h-step ahead predictor is

$$\hat{x}_{n+h} = \sum_{i=0}^{\infty} c_{i+h} \epsilon_{n-i},$$

and the MSE is $\mathrm{MSE}(\hat{x}_{n+h}) = \sigma^2 \sum_{i=0}^{h-1} c_i^2$. If ϵ_t is Gaussian, then the prediction bound for x_{n+h} is

$$\left(\hat{x}_{n+h} - z_{1-\alpha/2}\sigma \sqrt{\sum_{i=0}^{h-1} c_i^2}, \hat{x}_{n+h} + z_{1-\alpha/2}\sigma \sqrt{\sum_{i=0}^{h-1} c_i^2} \right).$$

Forecasting AR(p) process via MA

Let $x_1, \ldots, x_n$ be a time series from a causal AR(p) process $x_t = a_1 x_{t-1} + \ldots + a_p x_{t-p} + \epsilon_t$, which can be written in the MA form $x_j = \sum_{i=0}^{\infty} c_i \epsilon_{j-i}$, $c_0 = 1$.

From the previous slide, for $h = 1$, we have

$$\begin{aligned}
\hat{x}_{n+1} &= \sum_{i=0}^{\infty} c_{i+1} \epsilon_{n-i} = \sum_{j=1}^{\infty} c_j \epsilon_{n+1-j} \\
&= \sum_{j=1}^{\infty} c_j \epsilon_{n+1-j} + c_0 \epsilon_{n+1} - c_0 \epsilon_{n+1} = \sum_{j=0}^{\infty} c_j \epsilon_{n+1-j} - c_0 \epsilon_{n+1} \\
&= x_{n+1} - \epsilon_{n+1} = a_1 x_{n+1-1} + \ldots + a_p x_{n+1-p} + \epsilon_{n+1} - \epsilon_{n+1}.
\end{aligned}$$

In a similar way, we have

$$\hat{x}_{n+2} = a_1 \hat{x}_{n+1} + a_2 x_n + \ldots + a_p x_{n+2-p}.$$

Forecasting/predictor of AR process

Theorem. For a time series $x_1, \ldots, x_n$ from a causal AR(p) process $a(B)x_t = \epsilon_t$, which can be written as $x_j = \sum_{i=0}^{\infty} c_i \epsilon_{j-i}$, where ϵ_t is a white noise process with variance σ^2,
the h-step ahead predictor is

$$\hat{x}_{n+h} = \begin{cases} \sum_{j=1}^{h-1} a_j \hat{x}_{n-j+h} + \sum_{j=h}^{p} a_j x_{n-j+h} & h = 1, 2, \ldots, p \\ \sum_{j=1}^{p} a_j \hat{x}_{n-j+h} & h = p+1, p+2, \ldots, \end{cases}$$

and the MSE is $\mathrm{MSE}(\hat{x}_{n+h}) = \sigma^2 \sum_{i=0}^{h-1} c_i^2$.

Example on forecasting AR(2)

Task. Let 46,45,48,46 be a time series from an AR(2) process $x_t - 47 = 1.3(x_{t-1} - 47) - 0.7(x_{t-2} - 47) + \epsilon_t$. Compute the predictors $\hat{x}_5, \hat{x}_6, \hat{x}_7, \hat{x}_8$.

The given AR(2) process is causal since coefficients $a_1 = 1.3$ and $a_2 = -0.7$ satisfy $a_2 + a_1 < 1$, $a_2 - a_1 < 1$, $|a_2| < 1$, see slide 9 of Lecture 9.

Since the mean $\mu = 47$ is given, we find $y_1 = x_1 - 47 = -1$, $y_2 = 45 - 47 = -2$, $y_3 = 1$, $y_4 = -1$. Then we have

$$\hat{y}_5 = 1.3y_4 - 0.7y_3 = 1.3 \cdot (-1) - 0.7 \cdot 1 = -2,$$

$$\hat{y}_6 = 1.3\hat{y}_5 - 0.7y_4 = 1.3 \cdot (-2) - 0.7 \cdot (-1) = -1.9,$$

$$\hat{y}_7 = 1.3\hat{y}_6 - 0.7\hat{y}_5 = 1.3 \cdot (-1.9) - 0.7 \cdot (-2) = -1.07,$$

$$\hat{y}_8 = 1.3\hat{y}_7 - 0.7\hat{y}_6 = 1.3 \cdot (-1.07) - 0.7 \cdot (-1.9) = -0.06.$$

Finally, $\hat{x}_5 = \hat{y}_5 + 47 = 45, \hat{x}_6 = 45.1, \hat{x}_7 = 45.93, \hat{x}_8 = 46.94$.

R code for forecasting AR(2)

Task. Let 46,45,48,46 be a time series. Assume that an AR(2) process $x_t - 47 = 1.3(x_{t-1} - 47) - 0.7(x_{t-2} - 47) + \epsilon_t$ is appropriate for forecasting. Depict the time series and the predictors $\hat{x}_5, \ldots, \hat{x}_{40}$.

```
® x=c(46,45,48,46); n=length(x);
a=c(1.3,-0.7); p=length(a);
mu=47; h=40-n;
x=c(x,rep(0,h));
for(i in (1:h))
  x[n+i]=mu+sum(a*(x[seq(n+i-1,n+i-p,by=-1)]-mu));
plot((1:(n+h)),x,pch=19,main="time series (blue) and
forecast")
points((1:n),x[1:n],pch=19,col="blue")
```

Learning Lecture 12 with AI

Try these prompts.

Conceptual Mastery

Prompt: "Explain the intuition behind the h-step ahead predictor $\hat{x}_{n+h} = a^h x_n$ for an AR(1) process. Why does the forecast decay toward the mean as h increases?"

Algorithmic Practice

Prompt: "Walk me through the recursive forecasting algorithm for an AR(2) process. If $y_n = -1$ and $y_{n-1} = 1$ with $a_1 = 1.3, a_2 = -0.7$, show me how to calculate $\hat{y}_{n+1}$ and $\hat{y}_{n+2}$ step-by-step."

Coding with R

Task: Ask an AI to explain how the R loop on slide 12 handles indices to ensure it uses previous forecasts ($\hat{x}$) when the lag exceeds the available data points (n).